SE4041 MADD • ASSIGNMENT 02 (PART B: TVOS)

PartyPlay TV

Your Living Room Game Show

Native Apple tvOS Multiplayer Trivia Platform

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MODULE: SE4041 – Mobile Application Design & Development

DEGREE / FACULTY: BSc (Hons) in Information Technology • SLIIT

TECH STACK: tvOS 17.2 SDK • Swift 5.9 • Apple TV 4K

tvOS 17.2 | Focus Engine | CoreImage QR

100% Offline Demo | 10/10 Tests Passed



Living Room Broadcast Stage

Transforms the Apple TV 4K into an interactive game show host. Controllable via the physical Siri Remote or companion mobile buzzers.

PROBLEM STATEMENT

The Living Room Entertainment Disconnect

4K television screens are massive, yet shared living room play is missing.

THE SCREEN

Passive Video

4K TVs mostly play video

- Families sit together but watch passively
- The TV acts as a receiver, not a game host
- Little content invites shared doing and play

THE HABIT

Small-Screen Silos

Mobile trivia isolates players

- Everyone looks down into personal phone screens
- Kills the communal living-room atmosphere
- Lacks the thrill of a shared game show

THE SETUP

Costly & Complex

Console party games have friction

- Jackbox packs cost $30+ each
- Requires multiple gamepads or cloud servers
- No free, native offline Apple TV alternative

THE GAP: Turn the living room's centerpiece screen into an instant, social party game show.

PROPOSED SOLUTION

A Live Game Show on Your TV Screen

PartyPlay TV turns the Apple TV into an interactive game-show stage with zero setup friction.

THE MAIN IDEA: Turn the living-room TV into a broadcast game show you play with a remote.

1. NATIVE 10-FOOT FOCUS UI

Built for the couch with SwiftUI @FocusState, parallax card elevation (1.06x), and glowing cyan halos.

2. DUAL-MODE FLEXIBILITY

100% offline with smart AI bots for solo demo play, plus instant CoreImage QR codes for phone buzzers.

3. LIVE GAME ENGINE

Real-time countdown ring, dynamic +50% speed bonuses, instant educational feedback, and podium rankings.

SOLUTION ARCHITECTURE

▶ **HOST SCREEN (Apple TV 4K)**
Broadcasts questions, circular timer & podium

▶ **INTERACTION MODES**
Siri Remote Focus Engine • CoreImage QR Handshake

▶ **CONTESTANT ROSTER**
Solo Remote Host • Simulated AI Bots • Phone Buzzers

▶ **SWIFT MVVM CORE**
GameEngine (Timer/State) • ScoreService (Bonuses)

VALUE PROPOSITION & KEY FEATURES

Fast Setup, Fierce Competition

Instant couch entertainment designed for players of all ages.

10-FOOT DESIGN

Siri Remote First

- Parallax card lift (1.06x)
- Glowing cyan focus halos
- Readable from 3+ meters
- Tactile sound feedback

CURATED TRIVIA

6 Categories

- 42 verified questions
- Science, Gaming, Movies
- Sri Lanka Quiz heritage
- Easy, Medium, Hard tiers

REACTIVE ENGINE

Dynamic Scoring

- Circular countdown ring
- Up to +50% speed bonus
- Anti-spam submission lock
- Tie-breaker rank logic

ZERO FRICTION

100% Offline Ready

- Works with zero internet
- Simulated AI bots included
- Native CoreImage QR codes
- Zero accounts, zero ads

APPLICATION OVERVIEW & USER FLOW

One Clean Path Through 6 Screens

A smooth linear journey powered by SwiftUI NavigationStack and AppRoute routing.

STAGE 1
Home
Hero branding, Play Now & recent match summary.

STAGE 2
Categories
6 trivia themes with custom palettes & difficulty tags.

STAGE 3
Setup
Select timer (10-20s), question count & tier.

STAGE 4
Lobby
Room code PLAY88, QR code & AI bot roster.

STAGE 5
Quiz Arena
Countdown ring, 2x2 answers & instant reveal.

STAGE 6
Podium
Gold/Silver/Bronze pedestals & rankings table.

SIRI REMOTE CONTROL GESTURES

- **Touchpad Swipe:** Glide focus between category cards and answer choices with native spring physics.
- **Clickpad Select:** Confirm selection, submit answer choice, and advance questions.
- **Menu / Back Button:** Pop cleanly back up the navigation stack without losing game session state.

FEATURE SHOWCASE I

10-Foot Design & Siri Remote Focus

Television-calibrated layout, parallax card scaling, and tactile focus states.

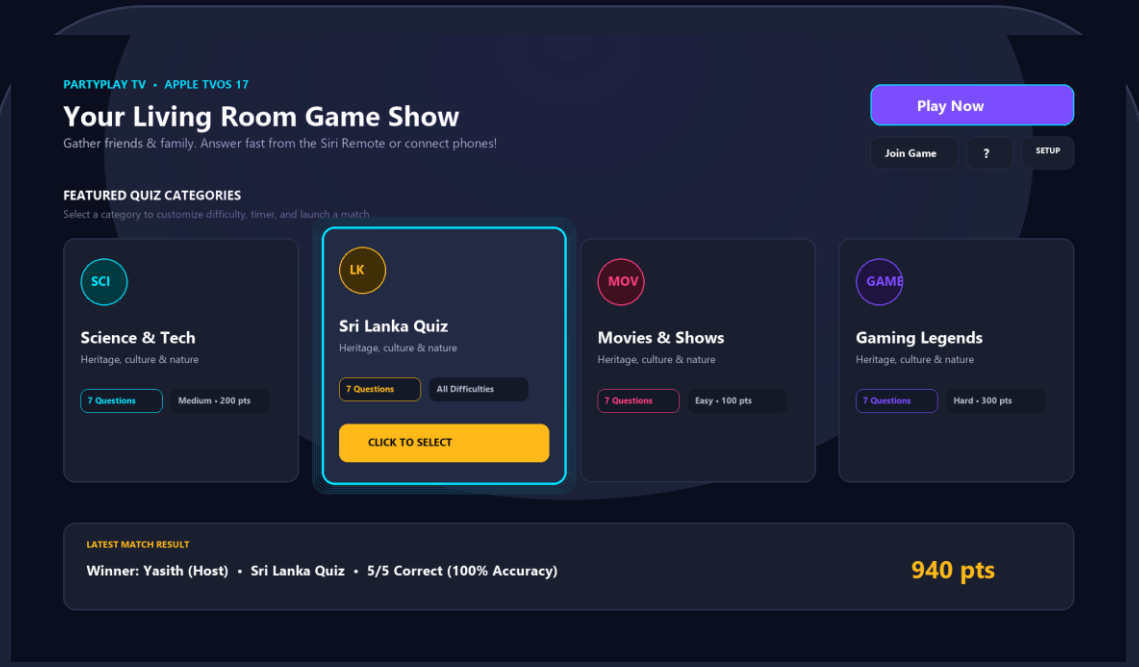
FOCUS ENGINE ARCHITECTURE

▶ Apple tvOS HIG Standard
90pt TV-safe margins with distance typography readable from 3+ meters away.

▶ @FocusState Management
Pure SwiftUI focus orchestration without mouse pointers or touchscreens.

▶ Parallax Card Elevation
Focused cards lift 1.06x with a bright cyan glowing border (#00E5FF).

▶ Sri Lanka Quiz Spotlight
Dedicated category featuring verified national history, geography, and sports.

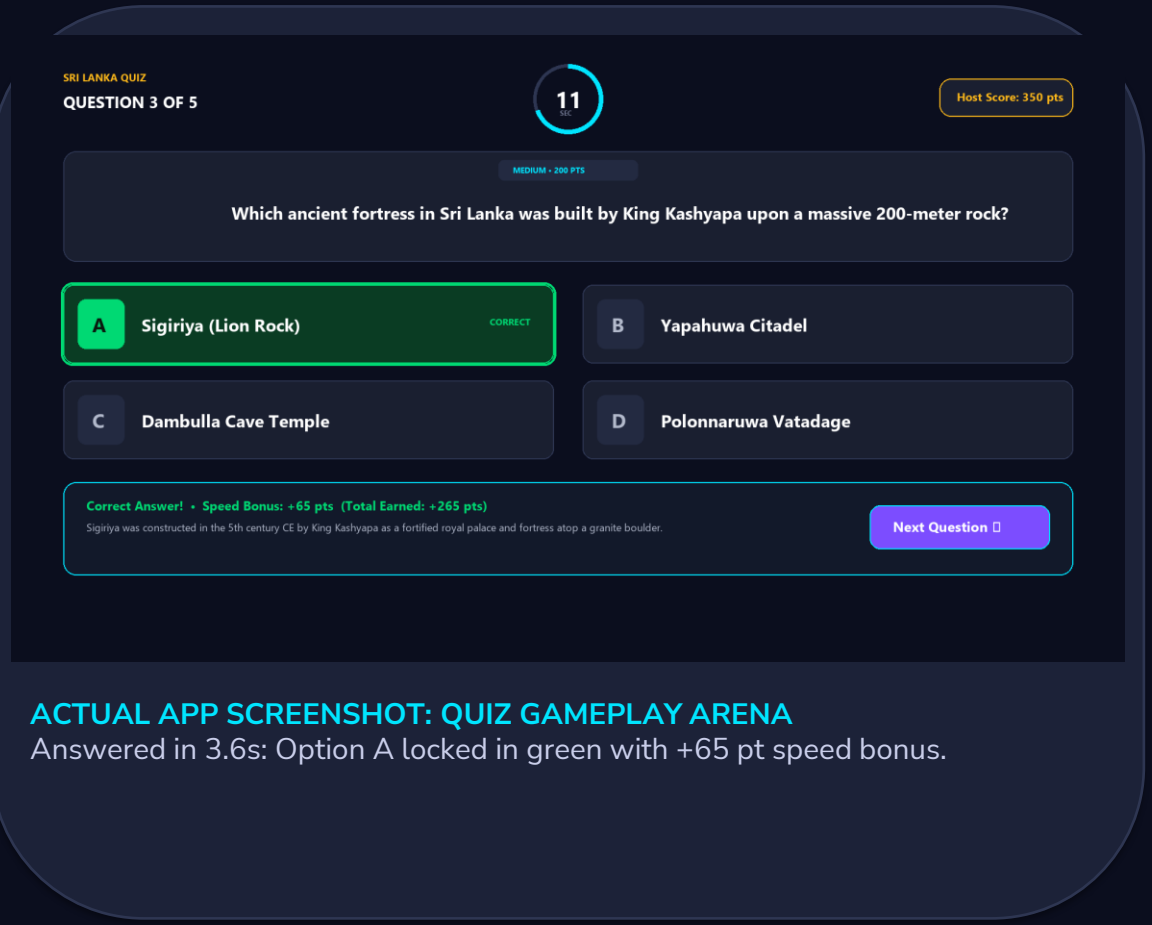


ACTUAL APP SCREENSHOT: HOME & CATEGORY SELECTION
Focused card lifts with parallax scale and cyan neon glow.

FEATURE SHOWCASE II

Real-Time Timer & Speed-Bonus Scoring

Broadcast question arena with synchronized countdown, speed bonuses, and educational feedback.



ACTUAL APP SCREENSHOT: QUIZ GAMEPLAY ARENA
Answered in 3.6s: Option A locked in green with +65 pt speed bonus.

SCORING & FEEDBACK MECHANICS

▶ Synchronized Countdown Ring
Color shifts Cyan (>50%) ➔ Amber (>20%) ➔ Red (<20%).

▶ Speed-Bonus Algorithm
Formula: Score = Base + (Base * 0.5 * TimeLeft / TotalTime).

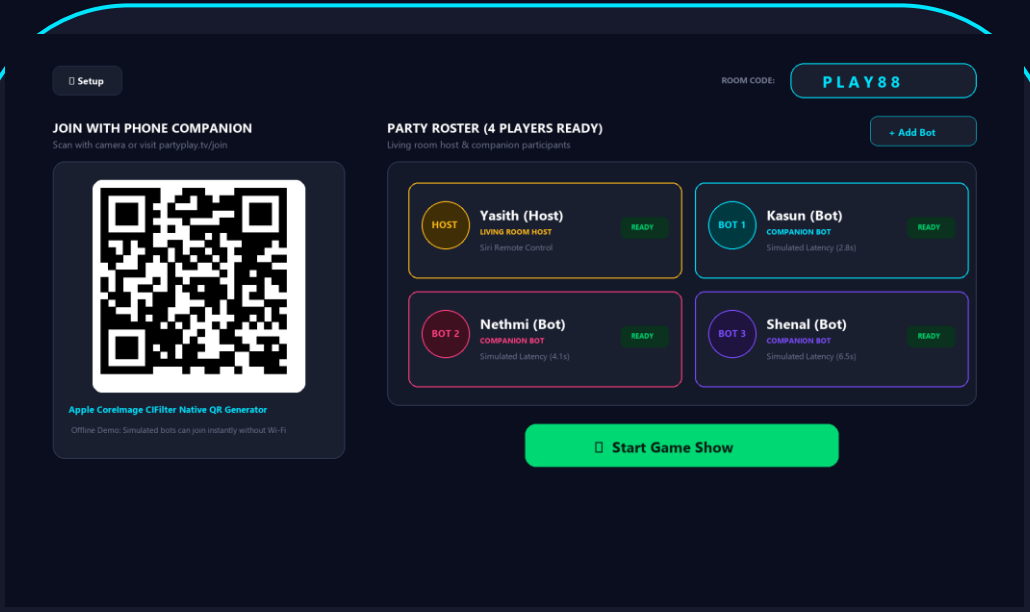
▶ Anti-Spam Input Lock
State machine locks on initial click to prevent double submissions.

▶ Instant Educational Reveal
Displays verified historical context and auto-focuses 'Next Question'.

FEATURE SHOWCASE III

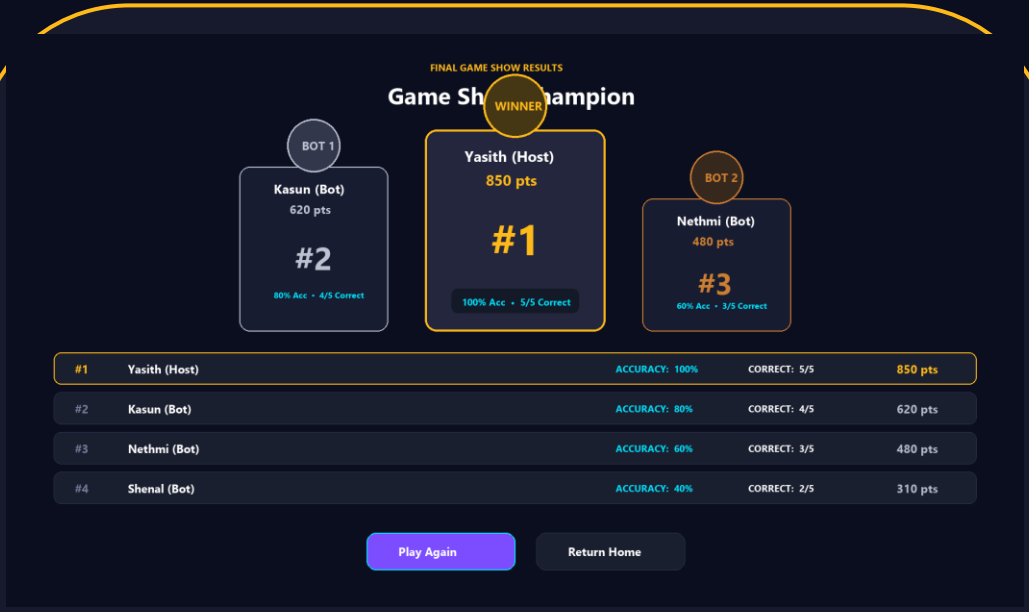
Simulated Bot Lobby & Winner Podium

Dual-mode multiplayer with native CoreImage QR pairing and deterministic podium rankings.



COREIMAGE QR & OFFLINE BOTS

- Native CoreImage QR: Encodes companion URLs without external libraries.
- Simulated AI Bots: Bots simulate human response delays (2–15s) for instant offline play.



CHAMPION PODIUM CELEBRATION

- 3D-Style Podium: Gold (#1, 250pt), Silver (#2, 190pt), Bronze (#3, 150pt).
- Deterministic Tie-Break: Score ➔ Correct Count ➔ Accuracy %.
- JSON Storage: Archives match history to Application Support.

TARGET AUDIENCE & MARKET

Capturing the Shared Living Room

Targeting multi-generational families, university social groups, and smart classrooms.

PRIMARY TARGET

Families

Wholesome shared screen time

- Grandparents & kids play together
- Safe, educational heritage trivia
- Replaces silent phone scrolling

SECONDARY TARGET

University Dorms

Fast-paced party gaming

- Social game nights & dorm tournaments
- Competitive speed-bonus tension
- Zero cost, no console required

TERTIARY TARGET

Classrooms

Interactive visual quizzes

- AirPlay quizzes on lecture displays
- High legibility from back of room
- Verified educational explanations

MARKET OPPORTUNITY & DATA METRICS

- **40M+ Active Apple TV Devices:** Premium living-room household base with 4K screens (Statista / Apple).
- **$18.2B Party Gaming Market:** Growing at 8.4% annually with high demand for co-located social play (Newzoo).
- **Blue Ocean Opportunity:** Zero direct competitors offering localized Sri Lankan trivia on Apple tvOS.

DIFFERENTIATION & ROADMAP

Why PartyPlay TV Wins & Future Roadmap

Clear advantages over existing alternatives, backed by a phased expansion plan.

Feature	PartyPlay TV	Jackbox	Kahoot!	TV Clones
Native 10-Foot UI	**Yes (Siri Remote)**	Port only	Mobile only	Clunky
Price / Barrier	**Free Prototype**	$30 / pack	$10+/mo	Ad-heavy
100% Offline	**Yes (Bots+JSON)**	Needs Net	Cloud only	Crashes
Local Heritage	**Yes (Sri Lanka)**	No	Unverified	None

CORE ADVANTAGES

- Zero Cost: Free prototype with no paywalls or ads
- Couch Ergonomics: First-class Siri Remote Focus Engine
- Offline Autonomy: Never fails when Wi-Fi is spotty
- Culturally Rich: First dedicated Sri Lankan trivia on tvOS

PHASED ROADMAP

PHASE 1: IMPLEMENTED (CURRENT)

- Native tvOS 17 app in Swift 5.9
- 6 categories with 42 verified questions
- Siri Remote focus & offline bot simulation
- 10/10 passing unit tests & JSON storage

PHASE 2: PROPOSED COMPANION (NEXT)

- Lightweight WebSocket server
- Real-time phone buzzers via partyplay.tv/join
- Zero app install needed on phones

PHASE 3: FUTURE ECOSYSTEM

- FaceTime SharePlay for remote game nights
- Custom trivia question creator portal
- Siri voice-command answer support

TECHNICAL IMPLEMENTATION

Native Architecture & Verified Quality

Clean MVVM architecture, 100% native Apple frameworks, and verified automated tests.

CORE ARCHITECTURE & TECH STACK

▶ Swift 5.9 & tvOS 17.2 SDK
Built in Xcode 15.2 with under 15MB binary size.

▶ Strict MVVM Pattern
Centralized AppContainer manages service dependency injection.

▶ Combine State Machine
Coordinates countdown timers, input locks, and score calculations.

▶ CoreImage Vector QR
Native CIFilter draws sharp QR codes on-the-fly without 3rd-party pods.

▶ 0 External Dependencies
100% native Apple frameworks for maximum stability and speed.

VERIFIED UNIT TESTING & METRICS

✓ 10/10 Passing Unit Tests (100% Pass Rate)
- ScoreServiceTests (4/4): Base points, speed bonuses, tie-breaks
- QuizDataServiceTests (4/4): JSON integrity, category filters, Sri Lanka content
- GameEngineTests (3/3): Clean reset, single-click lock, game loop

✓ Verified Performance
- Fluid 60 FPS remote focus navigation on Apple TV 4K
- Zero memory leaks across complete match cycles
- Full Reduce Motion & High Contrast support

✓ Practical Impact
- Makes television an active social medium for the living room
- Delivers cultural knowledge in a fun, competitive format

CONCLUSION & DEMONSTRATION

Bringing the Game Show to Every Living Room

A complete, production-ready tvOS platform ready for live simulator demonstration.

INNOVATION

Living Room Stage

Replaces passive video and solitary phone screens with a broadcast game show.

TECHNICAL RIGOR

Native MVVM & Tests

Swift 5.9, tvOS 17 SDK, 10/10 passing unit tests, and 0 third-party bloat.

ZERO BARRIERS

Instant Offline Play

100% offline out-of-the-box with smart AI bots and QR companion pairing.

PROCEEDING TO LIVE SIMULATOR DEMO ON APPLE TV 4K

Live Demonstration Path:

1. Home Stage: Hero game show branding & Siri Remote focus scaling
2. Category Select: Browse the 6 themes and select the 'Sri Lanka Quiz'
3. Party Lobby: Room code PLAY88, QR generator & simulated bots
4. Quiz Arena: Circular countdown ring, speed bonus (+50%) & educational context
5. Champion Podium: 3D-style podium celebration and accuracy rankings

Thank you for your time. Presenter: Jayasundara R.K.M.J.Y. (IT22136374) • SE4041 MADD Part B